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Differential Equation

i. $y = A\cos(ax - b) \quad -(1)$

$$\frac{dy}{dx} = A[-\sin(ax-b)] \cdot a$$

$$= -Aa\sin(ax-b)$$

$$\frac{d^2y}{dx^2} = -Aa\cos(ax-b)\, a$$

$$= -Aa^2\cos(ax-b)$$

$$= -a^2y \quad [\because \text{by } (1)]$$

$$\therefore \frac{d^2y}{dx^2} + a^2y = 0$$

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ii) $xy = pe^{x} + qe^{-x}$

$\Rightarrow x.\dfrac{dy}{dx} + y = pe^{x} - qe^{-x}$

$\Rightarrow x.\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} + \dfrac{dy}{dx} = pe^{x} + qe^{-x}$

$\Rightarrow x\dfrac{d^2y}{dx^2} + 2.\dfrac{dy}{dx} = xy$

$\Rightarrow x\dfrac{d^2y}{dx^2} + 2.\dfrac{dy}{dx} - xy = 0$

iii) $y = A\cos px + B\sin px$

$\dfrac{dy}{dx} = A - p\sin px + B.p\cos px$

again,

$\dfrac{d^2y}{dx^2} = -Ap[p\cos px] + B.p[-p\sin px]$

$= -Ap^2\cos px - Bp^2\sin px$

$= -p^2[A\cos px + B\sin px]$

$= -p^2y \quad \therefore \dfrac{d^2y}{dx^2} + p^2y = 0$

Variable Sperable

1. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

$\Rightarrow \frac{dx}{dy} = \frac{e^x}{e^y} + \frac{x^2}{e^y}$

$\Rightarrow e^y \cdot dy = (e^x + x^2)\, dx$

$\therefore$ Integration, $e^y = e^x + \frac{x^3}{3} + c$ Ans

2. $x(y-3)\, dy = 4y\, dx$

$\Rightarrow \frac{y-3}{4y}\, dy = \frac{1}{x}\, dx$

$\Rightarrow \frac{1}{4} - \frac{3}{4} \cdot \frac{1}{y} \cdot dy = \frac{1}{x} \cdot dx$

Now integrating,

$$\int \left(\frac{1}{4} - \frac{3}{4} \cdot \frac{1}{y}\right) dy = \int \frac{1}{x}\, dx$$

$\Rightarrow \frac{1}{4} \cdot y - \frac{3}{4} \ln y = \ln x + c$

$\therefore \frac{y}{4} - \frac{3}{4} \ln y = \ln x + c$ Ans

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3. $\frac{dy}{dx} = xy^3(1+x^2)^{-\frac{1}{2}}$

Given that,

$$\frac{dy}{dx} = xy^3(1+x^2)^{-\frac{1}{2}}$$

$$\Rightarrow \frac{1}{y^3}\,dy = x(1+x^2)^{-\frac{1}{2}}\,dx$$

$$\Rightarrow \frac{1}{y^3}\,dy = \frac{1}{2}(x+x^3)\,dx$$

Now integrating,

$$\int \frac{1}{y^3}\,dy = \int \frac{1}{2}(x+x^3)\,dx$$

$$\Rightarrow \int y^{-3}\,dy = \frac{1}{2}\int (x+x^3)\,dx$$

$$\Rightarrow \frac{y^{-3+1}}{-3+1} + c = \frac{1}{2}\left(\frac{x^2}{2} + \frac{x^4}{4}\right)$$

$$\Rightarrow \frac{y^{-2}}{-2} + c = \frac{2x^2+x^4}{4}$$

$$\Rightarrow -\frac{1}{y^2} + c = \frac{2x^2+x^4}{4}$$

$$\therefore -\frac{4}{y^2} + c = 2x^2+x^4$$

Ans

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Reducible Separable Variable

1. $\frac{dy}{dx} = (4x + y + 1)^2$

Soln:

$$\frac{dy}{dx} = (4x+y+1)^2 \quad \text{--- (1)}$$

Let, $4x + y + 1 = v$

$$\therefore 4 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{dv}{dx} - 4$$

from (1), $\frac{dv}{dx} - 4 = v^2$

$$\Rightarrow \frac{dv}{dx} = (v^2+4)$$

$$\therefore \int \frac{1}{v^2+2^2}\, dv = \int dx$$

$$\Rightarrow \frac{1}{2} \cdot \tan^{-1}\frac{v}{2} = x + c$$

$$\therefore \frac{1}{2}\tan^{-1}\left(\frac{4.x+y+1}{2}\right) = x + c$$

Ans

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2. $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$

Soln: Given that,

$$\frac{dy}{dx} = \sin(x+y) + \cos(x+y) \quad \text{---}(1)$$

Let,

$$x+y = v$$

$$\Rightarrow 1 + \frac{dy}{dx} = \frac{dv}{dx} \quad \text{or,} \quad \frac{dy}{dx} = \frac{dv}{dx} - 1$$

eqn (1) becomes,

$$\frac{dv}{dx} - 1 = \sin v + \cos v$$

$$\Rightarrow \frac{dv}{dx} = 1 + \sin v + \cos v$$

$$\Rightarrow dx = \frac{dv}{1 + \sin v + \cos v}$$

$$= \frac{dv}{2\cos^2\frac{v}{2} + 2\sin\frac{v}{2}\cos\frac{v}{2}}$$

$$= \frac{dv}{2\cos^2\frac{v}{2}\left(1+\tan\frac{v}{2}\right)}$$

$$dx = \frac{\frac{1}{2}\sec^2\frac{v}{2}\,dv}{1+\tan\frac{v}{2}}$$

Integrating.

$$\log\left(1+\tan\tfrac{1}{2}v\right) = x+c$$

$$\Rightarrow \log\left[1+\tan\tfrac{1}{2}(x+y)\right] = x+c$$

which is the required solution.

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Homogeneous Differential Equation

1. $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$

$\Rightarrow (x^2 - xy) \frac{dy}{dx} = -y^2$

$\Rightarrow \frac{dy}{dx} = \frac{y^2}{xy - x^2}$ —— (i)

Let, $y = vx$

$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$

From (i), $v + x \frac{dv}{dx} = \frac{v^2 x^2}{x \cdot vx - x^2}$

$\Rightarrow x \cdot \frac{dv}{dx} = \frac{v^2}{v-1} - v$

$\Rightarrow x \cdot \frac{dv}{dx} = \frac{v^2 - v^2 + v}{v-1}$

$\Rightarrow x \cdot \frac{dv}{dx} = \frac{v}{v-1}$

$\Rightarrow \int \frac{v-1}{v} \frac{dv}{dx} = \int \frac{1}{x} dx$

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$\Rightarrow \int \left(1-\frac{1}{v}\right) dv = \int \frac{1}{x} \cdot dx$

$\Rightarrow v - \ln v = \ln x + \ln c$

$\Rightarrow \ln e^v - \ln v = \ln (cx)$

$\Rightarrow \ln \frac{e^v}{v} = \ln (cx)$

$\Rightarrow \frac{e^v}{v} = cx$

$\Rightarrow e^v = vcx$

$\Rightarrow e^{y/x} = \frac{y}{x} cx \quad [y = vx]$

$\Rightarrow e^{y/x} = cy$ <u>Ans</u>

2. $\frac{dy}{dx} + \frac{y(x+y)}{x^2} = 0$

<u>Soln:</u> Given that,

$$\frac{dy}{dx} + \frac{y(x+y)}{x^2} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y(x+y)}{x^2} \quad \text{--- (1)}$$

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let,

$$y = vx$$

$$\frac{dy}{dx} = v + x \cdot \frac{dv}{dx}$$

eq^n (i) becomes,

$$v + x \cdot \frac{dv}{dx} = - \frac{vx(x+vx)}{x^2}$$

$$\Rightarrow v + x \cdot \frac{dv}{dx} = - \frac{vx^2 + v^2x^2}{x^2}$$

$$\Rightarrow x \cdot \frac{dv}{dx} = -v - v^2 - v$$

$$\Rightarrow x \cdot \frac{dv}{dx} = -2v - v^2$$

$$\Rightarrow \frac{1}{x} \cdot \frac{dx}{dv} = \frac{1}{-(v^2+2v)}$$

$$\Rightarrow \frac{dx}{x} = \frac{1}{-(2v+v^2)} dv$$

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$$\Rightarrow \frac{dx}{x} = \frac{1}{-(2v+v^2+1-1)} dv$$

$$\Rightarrow \frac{dx}{x} = -\frac{1}{(v+1)^2-1} dv$$

Integrating.

$$\ln x = -\frac{1}{2} \ln \left|\frac{v+1-1}{v+1+1}\right|$$

$$\Rightarrow \ln x = -\frac{1}{2} \ln \left|\frac{v}{v+2}\right|$$

$$\Rightarrow \ln x = -\frac{1}{2} \ln \left|\frac{y/x}{y/x+2}\right|$$

$$\Rightarrow \ln x = -\frac{1}{2} \ln \left|\frac{\frac{y}{x}}{\frac{y+2x}{x}}\right|$$

$$\Rightarrow \ln x = -\frac{1}{2} \ln \left|\frac{y}{x} \cdot \frac{x}{y+2x}\right|$$

$$\Rightarrow \ln x = -\frac{1}{2} \ln \left|\frac{y}{y+2x}\right|$$

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$\Rightarrow \ln x = \ln\left(\frac{y}{y+2x}\right)^{-\frac{1}{2}}$

$\therefore x = \left|\frac{y+2x}{y}\right|^{\frac{1}{2}} + c$

which is required eqⁿ.

Special case

1. Solve $(2x-2y+5)\frac{dy}{dx} = x-y+3$

Solⁿ: Given that,

$$(2x-2y+5)\frac{dy}{dx} = x-y+3$$

$$\Rightarrow \frac{dy}{dx} = \frac{x-y+3}{2x-2y+5}$$

$$= \frac{x-y+3}{2(x-y)+5} \quad \text{------ (i)}$$

Let, $x-y = v$

So that, $1-\frac{dy}{dx} = \frac{dv}{dx}$

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$$\frac{dy}{dx} = 1 - \frac{dv}{dx}$$

So, eqⁿ ① becomes

$$1 - \frac{dv}{dx} = \frac{v+3}{2v+5}$$

$$\Rightarrow \frac{dv}{dx} = 1 - \frac{v+3}{2v+5}$$

$$\Rightarrow \frac{dv}{dx} = \frac{2v+5-v-3}{2v+5}$$

$$\Rightarrow \frac{dx}{dv} = \frac{2v+5}{v+2}$$

$$\Rightarrow dx = \left\{ \frac{2(v+2)+1}{(v+2)} \right\} dv$$

$$= \left(2 + \frac{1}{v+2}\right) dv$$

Integrating,

$$x + c = 2v + \ln(v+2)$$

$$\Rightarrow x + c = 2(x-y) + \ln(x-y+2)$$

$$\Rightarrow x + c = x - 2y + \ln(x-y+2)$$

$$\Rightarrow 2y - x + c = \ln(x-y+2)$$

2. Solve $\frac{dy}{dx} = \frac{6x-4y+3}{3x-2y+1}$

Soln:

Given that,

$$\frac{dy}{dx} = \frac{6x-4y+3}{3x-2y+1}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2(3x-2y)+3}{3x-2y+1} \quad \text{---} (1)$$

Let,

$$3x-2y = v$$

$$\therefore \frac{dv}{dx} = 3-2 \cdot \frac{2v+3}{v+1}$$

$$\Rightarrow \frac{dv}{dx} = \frac{3v+3-4v-6}{v+1}$$

$$\Rightarrow \frac{dv}{dx} = -\frac{v+3}{v+1}$$

$$\Rightarrow dx = -\left(\frac{v+1}{v+3}\right) dv$$

Integrating.

$$\int dx = \int -\frac{(v+3)-2}{v+3}\, dv$$

$$x = -\int \frac{v+3}{v+3}\, dv - 2\int \frac{1}{v+3}\, dv$$

$$\Rightarrow x = -v + 2\log(v+3) + c$$

$$\Rightarrow x = 2y - 3x + 2\log(3x-2y+3) + c$$

$$\Rightarrow 4x - 2y = 2\log(3x-2y+3) + c$$

$$\Rightarrow 2x - y = \log(3x-2y+3) + \frac{1}{2}c$$

<u>Particular case</u>

1. $\frac{dy}{dx} = \frac{2x-y+1}{x+2y-3}$

<u>Soln</u> Given that,

$$\frac{dy}{dx} = \frac{2x-y+1}{x+2y-3}$$

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$$\Rightarrow (x+2y-3)\,dy = (2x-y+1)\,dx$$

$$\Rightarrow (x\,dy + y\,dx) + (2y-3)\,dy - (2x+1)\,dx = 0$$

Now integrating.

$$xy + \frac{1}{2}2y^2 - 3y - \frac{1}{2}2x^2 - x = 0$$

$$\Rightarrow xy + y^2 - 3y - x^2 - x = c$$

whic is required soln.

2. Solve $\dfrac{dy}{dx} + \dfrac{ax+hy+g}{hx+by+f} = 0$

<u>Soln:</u> Given that,

$$\frac{dy}{dx} + \frac{ax+hy+g}{hx+by+f} = 0$$

$$\Rightarrow (hx+by+f)\,dy + (ax+hy+g)\,dx = 0$$

$$\Rightarrow h(x\,dy + y\,dx) + (by+f)\,dy + (ax+g)\,dx = 0$$

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Integrating.

$$hxy + \frac{1}{2}by^2 + fy + \frac{1}{2}ax^2 + gx = c$$

$$\Rightarrow \quad ax^2 + 2hxy + by^2 + 2fy + 2gx + c = 0$$

which is required soln.

Linear D.E

1. $(1-x^2)\frac{dy}{dx} - xy = 1$

$$\Rightarrow \quad \frac{dy}{dx} - \frac{x}{(1-x^2)}y = \frac{1}{1-x^2} \quad \text{---} (1)$$

I.F (Integrating function) $= e^{\int P\,dx}$

$$= e^{\int \frac{-x}{1-x^2}dx}$$

$$= e^{\frac{1}{2}\int \frac{-2x}{1-x^2}dx}$$

$$= e^{\frac{1}{2}\ln(1-x^2)}$$

$$= e^{\ln(1-x^2)^{\frac{1}{2}}}$$

$$= \sqrt{1-x^2}$$

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The solution of the LDE is,

$$y.\text{I.F} = \int Q.\text{I.F}\,dx$$

$$\Rightarrow y.\sqrt{1-x^2} = \int \frac{1}{1-x^2} \times \sqrt{1-x^2}\,dx$$

$$\Rightarrow y.\sqrt{1-x^2} = \int \frac{1}{\sqrt{1-x^2}}\,dx$$

$$\Rightarrow y.\sqrt{1-x^2} = \sin^{-1}x + c$$

Ans

2. Solve $y\log y\,dx + (x-\log y)\,dy = 0$

Soln.

Given that,

$$y\log y\,dx + (x-\log y)\,dy = 0$$

$$\Rightarrow \frac{y\log y\,dx}{y\log y\,dy} + \frac{(x-\log y)\,dy}{y\log y\,dy} = 0$$

$$\Rightarrow \frac{dx}{dy} + \frac{x-\log y}{y\log y} = 0$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{y\log y} - \frac{1}{y} = 0$$

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$$\Rightarrow \frac{dx}{dy} + \frac{x}{y\log y} = \frac{1}{y} \quad -(1)$$

Now. $I.F = e^{\int \frac{1}{y\log y} dy}$

$= e^{\int \frac{1/y}{\log y} dy}$

$= e^{\ln(\log y)}$

$= \log y$

The soln of L.D.E is,

$x . I.F = \int Q . I.F \, dy$

$\Rightarrow x \log y = \int \frac{1}{y} . \log y . dy$ | let, $\log y = t$ $\therefore \frac{1}{y} dy = dt$

$\Rightarrow x . \log y = \int t \, dt$

$= \frac{t^2}{2} + c$

$= \frac{1}{2} t^2 + c$

$\therefore x . \log y = \frac{1}{2} (\log y)^2 + c$

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Bernoulli's D.E

1. $\frac{dy}{dx} = x^3y^3 - xy$

$\Rightarrow \frac{dy}{dx} + xy = x^3y^3$ —(i) [B.D.E]

from (i) diving by y^3

$y^{-3}\frac{dy}{dx} + xy^{-2} = x^3$ —(ii)

Let, $y^{-2} = v$

$\Rightarrow -2y^{-3}\frac{dy}{dx} = \frac{dv}{dx}$

$\Rightarrow y^{-3}\frac{dy}{dx} = -\frac{1}{2}\frac{dv}{dx}$

So, eqn (2) becomes,

$-\frac{1}{2}\frac{dv}{dx} + xv = x^3$

$\Rightarrow \frac{dv}{dx} - 2vx = -2x^3$ —(3) which is linear

Losectil

So, $I.F = e^{\int -2x\,dx}$

$= e^{-2 \cdot \frac{x^2}{2}} = e^{-x^2}$

Multiplying (3) by e^{-x^2} we get,

$e^{-x^2} \cdot \frac{dv}{dx} = e^{-x^2} \cdot 2vx = e^{-x^2} \cdot (-2x^3)$

$\Rightarrow \frac{d}{dx}\left(e^{-x^2} \cdot v\right) = -2x^3 \cdot e^{-x^2}$

Integrating,

$e^{-x^2} \cdot v = \int -2x^3 \cdot e^{-x^2}\,dx$ | Let, $x^2 = z$ $\Rightarrow 2x\,dx = dz$

$e^{-z} \cdot v = \int -z e^{-z} \cdot dz$

$= -z \int e^{-z} dz - \int \left\{ \frac{d}{dz}(-z) \int e^{-z} dz \right\} dz$

$= z e^{-z} - \int e^{-z} dz$

$= z \cdot e^{-z} + e^{-z} + c$

$\Rightarrow v = z + 1 + ce^{z}$

$\Rightarrow y^{-2} = 1 + x^2 + ce^{x^2}$

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2. $x.\frac{dy}{dx}+y=x^4y^3$

Soln:

Given that.

$$x.\frac{dy}{dx}+y=x^4y^3$$

$$\therefore \frac{dy}{dx}+\frac{1}{x}.y=x^3y^3 \quad \text{—— (1)}$$

Now, dividing by y^3 we get,

$$y^{-3}\frac{dy}{dx}+\frac{1}{x}.y^{-2}=x^3 \quad \text{—— (2)}$$

let, $y^{-2}=v$

so that $-2y^{-3}\frac{dy}{dx}=\frac{dv}{dx}$

$$\Rightarrow -y^{-3}\frac{dy}{dx}=-\frac{1}{2}\frac{dv}{dx}$$

So eqn (2) becomes

$$-\frac{1}{2}\frac{dv}{dx}+\frac{1}{x}v=x^3$$

$$\Rightarrow \frac{dv}{dx}-2v.\frac{1}{x}=-2x^3 \quad \text{—— (3)}$$

which is linear

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so, $I.F = e^{\int -2 \frac{1}{x} dx}$

$= e^{\ln (x)^{-2}} = \frac{1}{x^2}$

Multiplying (9) by $\left(\frac{1}{x^2}\right)$ we get,

$$\frac{1}{x^2} \cdot \frac{dv}{dx} - \frac{2v}{x} \cdot \frac{1}{x^2} = -2x^3 \cdot \frac{1}{x^2}$$

$$\Rightarrow \frac{d}{dx}\left(\frac{1}{x^2} \cdot v\right) = -2x$$

Integrating,

$$\frac{v}{x^2} = \int -2x\,dx$$

$$= -2\frac{x^2}{2} + c$$

$$\Rightarrow v = -x^2 \cdot x^2 + c$$

$$\therefore y^{-2} = -x^4 + c$$

3. $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

Soln:

Given that,

$$xy - \frac{dy}{dx} = y^3 e^{-x^2}$$

$$\Rightarrow \frac{dy}{dx} - xy = -y^3 e^{-x^2}$$

$$\Rightarrow y^{-3}\frac{dy}{dx} - xy^{-2} = -e^{-x^2} \quad \text{—— (i)}$$

Let $-y^{-3} = v$

$\Rightarrow -2.y$

$$\Rightarrow y^{-3}\frac{dy}{dx} = -\frac{1}{2}\frac{dv}{dx}$$

So, eqn (i) becomes,

$$-\frac{1}{2}\frac{dv}{dx} - xv = -e^{-x^2}$$

$$\frac{dv}{dx} + 2xy = 2e^{-x^2} \quad \text{—— (ii)}$$

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$I.F = e^{\int 2x\,dx}$

$= e^{x^2}$

Multiply eqn (ii) with e^x we get,

$e^{x^2}\frac{dv}{dx} + e^{x^2} 2xv = e^{x^2} \cdot 2e^{-x}$

$\Rightarrow \frac{d}{dx}(e^{x^2} \cdot v) = 2$

Integrating.

$e^{x^2} \cdot v = 2x$

$\Rightarrow e^{x^2} \cdot y^{-2} = 2x + c$

$\therefore y^{-2} = 2xe^{-x^2} + ce^{-x^2}$

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Exact D.E

1. $(x-2e^y)\,dy + (y + x\sin x)\,dx = 0$

Soln:

$\Rightarrow (y + x\sin x)\,dx + (x - 2e^y)\,dy = 0$ —— (i)

$M = y + \sin x$

$\frac{dM}{dy} = 1$ | $N = x - 2e^y$, $\frac{dN}{dx} = 1$

$\frac{dM}{dy} = \frac{dN}{dx}$ so It is exact

Integrating it with respect to x keeping as constant

$\int (y + x\sin x)\,dx = \int y.\,dx + \int x \sin x\,dx$

$= y.x + x\int \sin x.\,dx - \int\{\frac{d}{dx}x \int \sin x.\,dx\}\,dx$

$= xy - x\cos x + \sin x + 4$

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Again x free to term. N is $-2e^y$ so integration $-2e^y$ w.r.to y

$$\int -2e^y dy = -2\int e^y dy$$

$$= -2e^y + c_2$$

General soln.

$$y - x\cos x + \sin x - 2e^y = c$$

2. Solve $(y^4 + 4x^3y + 3x)dx + (x^4 + 4xy^3 + y + 1)dy = 0$

soln,

Given that,

$$(y^4 + 4x^3y + 3x)dx + (x^4 + 4xy^3 + y + 1)dy = 0 \quad \text{—(1)}$$

Comparing eqn (1) with $Mdx + Ndy = 0$ we get

$M = y^4 + 4x^3y + 3x$	$N = x^4 + 4xy^3 + y + 1$
$\Rightarrow \frac{\partial M}{\partial y} = 4y^3 + 4x^3$	$\Rightarrow \frac{\partial N}{\partial x} = 4x^3 + 4y^3$

Losecti

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As $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so the given eqn is exact.

Now, Integrating M w.r.to x keeping y as constant.

$$\int (y^4 + 4x^3y + 3x)\,dx = y^4x + 4y\cdot\frac{x^4}{4} + 3\cdot\frac{x^2}{2} + c_1$$

$$= xy^4 + x^4y + \frac{3}{2}x^2 + c_1$$

Again, x free term in N is y+1, so integrating y+1 w.r.to y,

we get, $\int (y+1)\,dy = \frac{y^2}{2} + y$

$$= \frac{1}{2}y^2 + y + c_2$$

Hence, the general soln is,

$$xy^4 + x^4y + \frac{3}{2}x^2 + \frac{1}{2}y^2 + y = c$$

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If not exact,

1. $(x^2+y^2)\,dx - 2xy\,dy = 0$

<u>soln,</u>

Given that,

$$(x^2+y^2)\,dx - 2xy\,dy = 0 \quad \text{--- (1)}$$

comparing (1) with $M\,dx + N\,dy = 0$ we get,

$M = x^2+y^2$ | $N = -2xy$

$\frac{\partial M}{\partial y} = 2y$ | $\frac{\partial N}{\partial x} = -2y$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ so the given eqn is not equal

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y+2y}{-2xy} = -\frac{2}{x}$$

$$I.F = e^{\int -2/x\,dx} = x^{-2} = \frac{1}{x^2}$$

Now multiplying ① by $\frac{1}{x^2}$ we get.

$$\frac{1}{x^2}(x^2+y^2)\,dx - 2xy\,\frac{1}{x^2}\,dy = 0$$

$$\Rightarrow \left(1+\frac{y^2}{x^2}\right)dx - \frac{2y}{x}\,dy = 0$$ which is now exact

Integrating $\left(1+\frac{y^2}{x^2}\right)$ w.r.to x as keeping y constant we get.

$$\int\left(1+\frac{y^2}{x^2}\right)dx = x^2+y^2\left(\frac{1}{x}\right)+c_1$$

$$= x - \frac{y^2}{x} + c_1$$

Again there is no x term is $\frac{-2y}{x}$ hence the general eqn is,

$$x^2 - \frac{y^2}{x} = c$$

$\therefore\ x^2 - y^2 = c$ Ans

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2. Solve $(y^4+2y)\,dx + (xy^3+2y^4-4x)\,dy = 0$

<u>Soln:</u>

Given that,

$$(y^4+2y)\,dx + (xy^3+2y^4-4x)\,dy = 0 \quad \text{--- (1)}$$

Comparing eqn (1) with $M\,dx + N\,dy = 0$ we get -

$M = y^4 + 2y$	$N = (xy^3 + 2y^4 - 4x)$
$\Rightarrow \dfrac{\delta M}{\delta y} = 4y^3 + 2$	$\dfrac{\delta N}{\delta x} = y^3 - 4$

As, $\dfrac{\delta M}{\delta y} \neq \dfrac{\delta N}{\delta x}$, the given eqn is not exact.

For exactness

$$\frac{\dfrac{\delta M}{\delta y} - \dfrac{\delta N}{\delta x}}{M} = \frac{4y^3+2-(y^3-4)}{y^4+2y}$$

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$$= \frac{3y^3+6}{y^4+2y}$$

$$= \frac{3(y^3+2)}{y(y^3+2)}$$

$$= \frac{3}{y}$$

which is a function of y

$$\therefore I.F = e^{-\int \frac{3}{y} dy}$$

$$= e^{-3\log y}$$

$$= e^{\log y^{-3}}$$

$$= y^{-3} = \frac{1}{y^3}$$

Multiplying by $\frac{1}{y^3}$, the eqn becomes

$$\left(y + \frac{2}{y^2}\right) dx + \left(x + 2y - \frac{4x}{y^3}\right) dy = 0$$

which is now exact.

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Integrating y^4+2y w.r.to x. keeping y as constant, we have, $\int y^4+2y$

$$= yx + \frac{2}{y^2}x + C_1$$

$$x + 2y - \frac{4x}{y^3}$$ the x free term is $2y$,

so, integrating $2y$ w.r.to y we get,

$$\int 2y = y^2 + C_2$$

So, the sol^n is

$$xy + \frac{2y}{y^2} + y^2 = c$$

Ans

3. Solve $(2x\log x - xy)\,dy + 2y\,dx = 0$

Sol^n

Given that,

$$(2x\log x - xy)\,dy + 2y\,dx = 0$$

$$2y\,dx + (2x\log x - xy)\,dy = 0 \quad \text{--- (i)}$$

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Comparing eqn (i) with $Mdx + Ndy$ we get,

$M = 2y$ | $N = 2x\log x - xy$

$\frac{\partial M}{\partial y} = 2$ | $\frac{\partial N}{\partial x} = 2.x.\frac{1}{x} + \log x . 2 - y$

$= 2 + 2\log x - y$

As, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ so the given eqn is not exact.

for exactness,

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2 - 2 - 2\log x + y}{2x\log x - xy}$$

$$= \frac{-2\log x + y}{x(2\log x - y)}$$

$$= \frac{-(2\log x - y)}{x(2\log x - y)}$$

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$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = -\frac{1}{x},$$

$$I.F = e^{-\int \frac{1}{x} dx}$$

$$= e^{-\ln x}$$

$$= e^{\ln x^{-1}}$$

$$= x^{-1}$$

$$= \frac{1}{x}$$

Now, Multiply eqn ①

$$\frac{1}{x} 2y\, dx + \frac{1}{x} (2x \log x - xy)\, dy = 0$$

Now, Integrating $\frac{1}{x} 2y dx$ w.r.to. x keeping y as constant. we get,

$$\int \frac{1}{x} 2y\, dx = 2y \ln x$$

Again x free term in N is $-y$. Now Integrating w.r. to y we get,

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$$-\int y\,dy = -\frac{y^2}{2}$$

Hence the general soln is,

$$2y\ln x - \frac{y^2}{2} = c.$$

Higher order D.E with CC

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1. $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx} + 4.\frac{dy}{dx} - 2y = e^x + \cos x$

Soln:

Given that,

$$\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx} + 4\frac{dy}{dx} - 2y = e^x + \cos x$$

$$\Rightarrow (D^3 - 3D^2 + 4D)y - 2y = e^x + \cos x$$

$$\Rightarrow (D^3 - 3D^2 + 4D - 2)y = e^x + \cos x \quad \text{--- (1)}$$

Now for c.f consider,

$$(D^3 - 3D^2 + 4D - 2)y = 0 \quad \text{--- (2)}$$

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Let $y = e^{mx}$ be the final solⁿ of eqⁿ

②.

$\therefore Dy = me^{mx}$, $D^2y = m^2e^{mx}$ and $D^3y = m^3e^{mx}$.

So. eqⁿ ② becomes,

$$(m^3 - 3m^2 + 4m - 2)e^{mx} = 0$$

But $e^{mx} \neq 0$

$\therefore m^3 - 3m^2 + 4m - 2 = 0$ when is the A.E

$\Rightarrow m^3 - m^2 - 2m^2 - 2m + 2m - 2 = 0$

$\Rightarrow m^2(m-1) - 2m(m-1) + 2(m-1) = 0$

$\Rightarrow (m-1)(m^2 - 2m + 1) = 0$

$m = 1$ or, $m^2 - 2m + 1 = 0$

$$\Rightarrow m = \frac{-(-2) \pm \sqrt{(-2)^2 - 4.1.2}}{2 \cdot 1}$$

$$= \frac{2 \pm 2i}{2}$$

$$= 1 \pm i$$

Losecti

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Hence, the C.F $= c_1 e^x + e^x (c_2 \cos x + c_3 \sin x)$

Again,

$$P.I = \frac{1}{D^3 - 3D^2 + 4D - 2} (e^x + \cos x)$$

$$= \frac{1}{D^3 - 3D^2 + 4D - 2} e^x + \frac{1}{D^3 - 3D^2 + 4D - 2} \cos x$$

$$= \frac{x}{3D^2 - 6D + 4} e^x + \frac{1}{-1^2 . D - 3 . -1^2 + 4D - 2} \cos x$$

$$= \frac{x}{3(1)^2 - 6.1 + 4} e^x + \frac{1}{3D + 1} \cos x$$

$$= x e^x + \frac{1}{3D + 1} \cos x$$

$$= x e^x + \frac{3D - 1}{(3D + 1)(3D - 1)} \cos x$$

$$= x e^x + \frac{3D - 1}{9D^2 - 1} \cos x$$

$$= x e^x + \frac{3D - 1}{9(-1) - 1)} \cos x$$

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$$= xe^x + \frac{3D-1}{-10}\cos x$$

$$= xe^x - \frac{3}{10} D(\cos x) + \frac{1}{10}\cos x$$

$$= xe^x - \frac{3}{10}\frac{d}{dx}(\cos x) + \frac{1}{10}\cos x$$

$$= xe^x - \frac{3}{10}(-\sin x) + \frac{1}{10}\cos x$$

$$= xe^x + \frac{3}{10}\sin x + \frac{1}{10}\cos x$$

Hence the complete general sol^n,

$$y = C.F + P.I$$

$$y = c_1 e^x + e^x(c_2\cos x + c_3 \sin x) + xe^x + \frac{3}{10}\sin x$$

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2. Solve $\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = e^{-x} + 1 + \sin 2x$

Soln:

Given that,

$$\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = e^{-x} + 1 + \sin 2x$$

$$\Rightarrow (D^3 + 3D^2 + 3D + 1)\, y = e^{x} + 1 + \sin 2x \quad \text{--- (i)}$$

Now for C.F consider,

$$(D^3 + 3D^2 + 3D + 1)\, y = 0 \quad \text{--- (ii)}$$

Let, $y = e^{mx}$ be final soln of eqn (ii)

$\therefore Dy = me^{mx}$ $\quad D^2y = m^2e^{mx}$

$D^3y = m^3e^{mx}$

So, eqn (ii) becomes,

$$(m^3 + 3m^2 + 3m + 1)\, e^{mx} = 0$$

But $e^{mx} \neq 0$

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$m^3 + m^2 + 2m^2 + 2m + m + 1 = 0$

$\Rightarrow m^2(m+1) + 2m(m+1) + 1(m+1) = 0$

$\Rightarrow (m+1)(m^2+2m+1) = 0$ $\cancel{\Rightarrow (m+1)(m+1)(m+1) = 0}$

$(m+1)(m^2+m+m+1) = 0$

$m = -1$ $\Rightarrow (m+1)(m+1)(m+1) = 0$

$m = -1, -1, -1$

~~So, (C1 + C2x + C3x^2) e^-x~~

So, $c.f = c_1 e^{x} + e^{-1}.(c_1 \cos 2x + c_2 \sin 2x)$

Again, Particular Integral,

$$P.I = \frac{1}{D^3+3D^2+3D+1} e^{-x} + 1 + \sin 2x$$

$$= \frac{1}{D^3+3D^2+3D+1} e^{-x} + \frac{1}{D^3+3D^2+3D+1} e^{x.0} + \frac{1}{D^3+3D^2+3D+1} \sin 2x$$

Losectil

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$$= \frac{1}{3D^2+6D+3} e^{-x} + \frac{1}{1} + \frac{1}{-(2)^2 \cdot D+3-(2)^2+3D+1} \sin 2x$$

$$= \frac{x^2}{6D+6} e^{-x} + 1 + \frac{1}{-4D-2+3D+1} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 + \frac{1}{-(D+11)} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 - \frac{D-11}{D^2-(11)^2} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 - \frac{D-4}{-(2)^2-121} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 - \frac{D-4}{-125} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 - \frac{D-4}{-125} \cdot D \sin 2x + \frac{4}{125} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 + \frac{1}{125} \frac{d}{dx} \sin 2x + \frac{4}{125} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 + \frac{1}{125} \times 2 \times \cos 2x + \frac{4}{125} \sin 2x$$

$$= \frac{x^3}{6} e^{-x} + 1 + \frac{2}{125} \cos 2x + \frac{11}{125} \sin 2x$$